

Rami I. Hanna

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Boston, MA

PROFILE

Robotics and autonomous systems engineer with an M.Sc. in Autonomous Systems from DTU and a Summa Cum Laude B.S. in Electromechanical Engineering from Wentworth. Across **PerPlant**, **Harvard Microrobotics**, **Raytheon Technologies**, and **DEKA R&D**, I have built and integrated robotics software, sensing, machine vision, controls, embedded interfaces, test systems, and mechatronic hardware. I am most interested in intelligent physical systems where software, learning, sensing, controls, and hardware have to work together reliably in the real world.

SKILLS

Programming: Python, C++, Embedded C, SQL; NumPy, SciPy, pandas/Polars, scikit-learn

ML / Vision: PyTorch, OpenCV, TensorRT, YOLOv8, UMAP, HDBSCAN

Robotics / Sensing: ROS2, MicroROS, NVIDIA Jetson; thermal, multispectral & RGB cameras; GPS/GNSS, IMU, LiDAR

Controls / Embedded: PID, LabVIEW, Modbus TCP/IP, Arduino, ESP32, motor/encoder control

Systems / Build: Linux/Ubuntu, CMake, Git, Docker, GitHub Actions, automated testing

Mechanical / Automation: SOLIDWORKS (CSWA), Fusion 360, CAD/FDM prototyping, fixture design, PLCs, UR5e, Keyence machine vision

PROFESSIONAL EXPERIENCE

PerPlant | Robotics Co-op

02/25 - 11/25 | Copenhagen, Denmark

- Integrated thermal-camera capture and GPS metadata into ROS2 C++/Python field workflows; used NVIDIA Jetson-based device platforms, contributed to production sensor-fusion work, and separately used TensorRT.
- Redesigned the multi-camera fixture and internal components in Fusion 360 to integrate a thermal camera alongside existing multispectral and RGB cameras.
- Built Python/CV data-curation tooling using detector embeddings, ROI filtering, UMAP/HDBSCAN, visual review, and grouped splitting to select ~50 representative images per batch from a 150,000+ image field dataset.

Harvard Microrobotics | Robotics Co-op

09/22 - 03/23 | Allston, MA

- Developed Flask/socket robot-operation interfaces spanning Embedded C and MicroROS to support teleoperated and autonomous underwater-robot workflows; worked on NVIDIA Jetson-based robot platforms.
- Contributed to robot communications, planetary-gearbox design, and subsystem integration, debugging issues across software, embedded hardware, and mechanical boundaries.

Raytheon Technologies | Innovation / Robotics Co-op

01/22 - 05/22 | Andover, MA

- Integrated PLCs, a Universal Robots UR5e, Keyence machine vision, and custom tooling for vision-guided robotic epoxy-bonding automation.
- Designed modular CAD fixtures for multiple circuit-card and surface-mount configurations to support repeatable alignment and robot access.

DEKA Research and Development | Controls Intern

04/19 - 09/19; 06/20 - 07/21 | Manchester, NH

- Implemented PID-controlled heating/cooling in LabVIEW over Modbus TCP/IP and supported electrical, impact, and thermal biomedical-device test workflows.
- Built RFID/Python/SQL traceability tooling and an Arduino/Python multi-source serial-data bench to structure engineering records and automate test-data capture.

SELECTED PROJECTS & RESEARCH

M.Sc. Thesis | Wearable Tendon-Driven Finger Actuator

03/26 - 08/26 | DTU

- Developed and experimentally evaluated a tendon-driven wearable finger-actuation system, using reduced-order tendon-routing models to guide mechanical redesign while integrating CAD/FDM hardware, closed-loop motor/encoder control, and Python/OpenCV load-side measurement.
- Demonstrated repeatable rigid-fixture operation across 1,000/1,000 commanded cycles with ~0.26% mean period error; comparison of encoder-side and camera-side motion helped identify compliance, slack, and load-path state as key limits to wearable transfer.

Teradyne Cartesian Robot & Tool-Changer | Published Robotic Test System 2023 | Wentworth / Teradyne

- Built and programmed a Raspberry Pi/ESP32 Cartesian robot with load-cell feedback, closed-loop load detection, scheduling/tool-change logic, and automatic tool changing to automate coaxial-connector mating and structured test-data collection.
- Achieved 0.01 mm travel precision and 0.5 g load standard deviation while automating repeatable connector mating and structured measurement across test cycles.

Football Video Analysis | Computer-Vision Prototype

2023 | Independent

- Built an end-to-end football video-analysis workflow: extracted broadcast frames, created and labelled a custom CVAT dataset, trained and compared YOLOv8 player/ball detectors, and integrated inference into a PySide review interface.
- Used the review workflow to inspect detector behavior under camera motion, occlusion, partial pitch visibility, and small-object ball detection, surfacing practical limitations of inference on broadcast footage.

SunnySips | Released iOS / Data Product

2025 - Present | Independent

- Built and released a SwiftUI iOS / FastAPI product combining location, weather, solar position, seating, and urban-occlusion data into time-specific sunny-cafe recommendations.
- Expanded venue and urban-occlusion coverage to Boston alongside Copenhagen, with Dockerized services, hosted snapshot delivery, caching/fallback behavior, validation, and GitHub Actions-backed release automation.

Categorical Diffusion for Mario Tile Grids | Structured Generative Modeling

2025 | DTU

- Explored categorical diffusion on 13-class one-hot Mario tile grids with a PyTorch U-Net/DDPM workflow and analyzed why FID alone was insufficient for structured/playability validity.

EDUCATION

Technical University of Denmark (DTU)

2024 - 2026 | Lyngby, Denmark

M.Sc. Autonomous Systems

Thesis: Modeling and Experimental Evaluation of a Wearable Tendon-Driven Finger Actuator for Neurorehabilitation Tasks

University of New South Wales (UNSW Sydney)

2025 | Sydney, Australia

Exchange Studies, Computer Science

Wentworth Institute of Technology

2019 - 2023 | Boston, MA

B.S. Electromechanical Engineering | Summa Cum Laude

Dean's Award in Electromechanical Engineering

SELECTED PUBLICATION

Increasing Efficiency and Reliability of RF Machinery Testing Using Cartesian Robotics and Automatic Data Collection. IEEE/SICE International Symposium on System Integration (SII), 2024, pp. 363-368.

DOI: [10.1109/SII58957.2024.10417202](https://doi.org/10.1109/SII58957.2024.10417202)

INTERESTS

Salsa; cycling & climbing/outdoors; cooking; building & tinkering.